

APSPDCL · COMMERCIAL ROOFTOP

950 KWP GRID-TIED SOLAR PLANT

HT CATEGORY I / II AT 11 KV

30-YEAR ANALYSIS

Net investment ₹3.44 Cr after input tax credit, generating 14,56,350 units in year one against an energy charge of ₹7.65 per unit.

PAYBACK 3.5 yrs Post-tax, undiscounted	PROJECT IRR 29.0% Unlevered, 30 years	COST OF SOLAR ENERGY ₹2.98 per kWh, against ₹7.65 from the grid	NET PRESENT VALUE ₹6.94 Cr Discounted at 10%
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PLANT AND GENERATION	INVESTMENT AND TAX BASIS																																
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ACCELERATED DEPRECIATION — FIRST 12 YEARS

YEAR	OPENING WDV (₹)	DEPRECIATION AT 40% (₹)	CLOSING WDV (₹)	TAX SHIELD AT 25.168% (₹)
1	3,43,80,952	1,37,52,381	2,06,28,571	34,61,199
2	2,06,28,572	82,51,429	1,23,77,143	20,76,720
3	1,23,77,143	49,50,857	74,26,286	12,46,032
4	74,26,285	29,70,514	44,55,771	7,47,619
5	44,55,772	17,82,309	26,73,463	4,48,572
6	26,73,463	10,69,385	16,04,078	2,69,143
7	16,04,078	6,41,631	9,62,447	1,61,486
8	9,62,447	3,84,979	5,77,468	96,892
9	5,77,468	2,30,987	3,46,481	58,135
10	3,46,480	1,38,592	2,07,888	34,881
11	2,07,888	83,155	1,24,733	20,928
12	1,24,733	49,893	74,840	12,557
Tax shield, years 1-12				86,34,162

COMMERCIAL SOLAR — FINANCIAL ANALYSIS

30-year post-tax cash flow

950 kWp
PLANT CAPACITY

How this table works. Electricity savings reduce an operating expense, so they increase taxable profit and are taxed at 25.168%. Depreciation reduces the same taxable profit. Both are applied to one figure, so the tax column is a net inflow in the early years when depreciation exceeds savings — shown in green. Net cash flow is therefore gross saving, less maintenance, less that tax. The shaded row is the year the investment is recovered.

YEAR	GENERATION (KWH)	GROSS SAVING (₹)	AMC (₹)	DEPRECIATION (₹)	TAX (₹)	NET CASH FLOW (₹)	CUMULATIVE (₹)
1	14,56,350	1,04,27,466	—	1,37,52,381	(8,36,815)	1,12,64,281	1,12,64,281
2	14,49,068	1,07,90,342	5,70,000	82,51,429	4,95,536	97,24,806	2,09,89,087
3	14,41,823	1,11,65,846	5,70,000	49,50,857	14,20,731	91,75,115	3,01,64,202
4	14,34,614	1,15,54,417	5,70,000	29,70,514	20,16,939	89,67,478	3,91,31,680
5	14,27,441	1,19,56,511	5,70,000	17,82,309	24,17,186	89,69,325	4,81,01,005
6	14,20,304	1,23,72,597	5,70,000	10,69,385	27,01,335	91,01,263	5,72,02,268
7	14,13,202	1,28,03,164	5,70,000	6,41,631	29,17,357	93,15,807	6,65,18,075
8	14,06,136	1,32,48,714	5,70,000	3,84,979	30,94,087	95,84,627	7,61,02,702
9	13,99,105	1,37,09,769	5,70,000	2,30,987	32,48,882	98,90,887	8,59,93,589
10	13,92,110	1,41,86,869	5,70,000	1,38,592	33,92,213	1,02,24,656	9,62,18,245
11	13,85,149	1,46,80,572	5,70,000	83,155	35,30,420	1,05,80,152	10,67,98,397
12	13,78,223	1,51,91,456	5,70,000	49,893	36,67,371	1,09,54,085	11,77,52,482
13	13,71,332	1,57,20,119	5,70,000	29,936	38,05,448	1,13,44,671	12,90,97,153
14	13,64,476	1,62,67,179	5,70,000	17,962	39,46,145	1,17,51,033	14,08,48,186
15	13,57,653	1,68,33,277	5,70,000	10,777	40,90,429	1,21,72,848	15,30,21,034
16	13,50,865	1,74,19,075	5,70,000	6,466	42,38,948	1,26,10,127	16,56,31,161
17	13,44,111	1,80,25,259	5,70,000	3,880	43,92,163	1,30,63,096	17,86,94,257
18	13,37,390	1,86,52,538	5,70,000	2,328	45,50,427	1,35,32,110	19,22,26,367
19	13,30,703	1,93,01,646	5,70,000	1,397	47,14,029	1,40,17,617	20,62,43,984
20	13,24,050	1,99,73,343	5,70,000	838	48,83,222	1,45,20,121	22,07,64,105
21	13,17,429	2,06,68,415	5,70,000	503	50,58,243	1,50,40,173	23,58,04,278
22	13,10,842	2,13,87,676	5,70,000	302	52,39,317	1,55,78,359	25,13,82,637
23	13,04,288	2,21,31,967	5,70,000	181	54,26,670	1,61,35,297	26,75,17,934
24	12,97,767	2,29,02,160	5,70,000	109	56,20,531	1,67,11,629	28,42,29,563
25	12,91,278	2,36,99,155	5,70,000	65	58,21,129	1,73,08,026	30,15,37,589
26	12,84,821	2,45,23,886	5,70,000	39	60,28,704	1,79,25,182	31,94,62,771
27	12,78,397	2,53,77,317	5,70,000	23	62,43,500	1,85,63,817	33,80,26,588
28	12,72,005	2,62,60,448	5,70,000	14	64,65,768	1,92,24,679	35,72,51,267
29	12,65,645	2,71,74,311	5,70,000	8	66,95,771	1,99,08,540	37,71,59,807
30	12,59,317	2,81,19,977	5,70,000	5	69,33,777	2,06,16,200	39,77,76,007
Total	4,06,65,894	53,65,25,471	1,65,30,000	3,43,80,945	12,22,19,463	39,77,76,007	39,77,76,007

COMMERCIAL SOLAR — FINANCIAL ANALYSIS

Return metrics, assumptions and basis

950 kWp
PLANT CAPACITY

RETURN METRICS

Simple payback	3.5 yrs
Discounted payback	4.5 yrs
Project IRR, unlevered	29.0%
Net present value at 10%	₹6.94 Cr

OVER THE FULL TERM

Levelised cost of energy	₹2.98 / kWh
Cumulative return, 30 years	1057%
Total net cash flow, after tax	₹39.78 Cr
30-year generation	4,06,65,896 kWh

COST OF A UNIT — SOLAR AGAINST THE GRID

Solar, levelised over 30 years		₹2.98
Grid energy charge today		₹7.65

Every unit the plant generates costs **₹2.98** across its life against **₹7.65** bought from APSPDCL today — and the grid figure reaches **₹23.86** by year 30 at 4% escalation, while the solar figure is fixed at the day of purchase. Over 30 years the plant generates 4,06,65,896 units, avoiding 33,346.0 tonnes of CO₂.

ASSUMPTIONS

- Tariff:** HT Category I / II at 11 kV at ₹7.65/kWh. Energy charge; ₹475/kVA/month demand charge excluded. APERC Retail Supply Tariff Order FY2025-26, continued for FY2026-27.
- Demand charges:** Excluded. Solar does not reduce contracted demand, so fixed and demand charges are unchanged by the plant.
- Tariff escalation:** 4% per year, applied to both the energy charge and the export rate.
- Generation:** 1,533 kWh per kWp in year one, degrading 0.5% per year.
- Unit split:** 90% self-consumed, 10% exported at ₹2.75/kWh. The plant is assumed to be sized to the site's daytime load.
- GST:** 5%, recovered in full as input tax credit. Figures are stated ex-GST. A buyer who cannot claim ITC should read the cost as ₹3.61 Cr.
- Depreciation:** 40% written-down value, Income Tax Act Appendix I. No additional depreciation under s.32(1)(ia), which s.115BAA forfeits.
- Tax:** 25.168% — s.115BAA at 22% plus 10% surcharge and 4% cess. Applied to savings and to depreciation alike.
- Loss set-off:** Early-year depreciation exceeds savings. The resulting loss is assumed to shelter the buyer's other business income in the same year.
- Commissioning:** Full first-year depreciation assumed, i.e. the plant is put to use on or before 30 September. Later commissioning halves the year-one charge, with the balance carried forward.
- Maintenance:** ₹600 per kWp per year from year 2, held flat. Year 1 is covered by the build.
- Inverter replacement:** Not modelled. A mid-life inverter replacement would reduce the figures shown.
- Financing:** None. All returns are unlevered — the return on the money invested, before any loan.
- Horizon:** 30 years, which runs beyond the module performance warranty.
- Discount rate:** 10% for net present value, discounted payback and levelised cost.
- Emissions:** Grid emission factor 0.82 kg CO₂ per kWh; tree equivalence at 45 trees per tonne is illustrative, not measured sequestration.

BASIS AND LIMITATIONS

This is an indicative financial analysis for a standard 950 kWp commercial rooftop plant. **It is not a quotation, an offer, or tax advice.** Pricing is subject to a site survey, roof condition, structure height, cable routing, evacuation distance and the APSPDCL sanction actually granted.

Tariffs, the depreciation block and the corporate tax rate are as we understand them at the date printed below and are subject to change by APERC and by the Finance Act. The tax treatment shown assumes a GST-registered company that has opted into s.115BAA and has sufficient other business income to absorb the early-year depreciation. **Have your chartered accountant confirm the tax position before you rely on it.**

Generation depends on site irradiance, shading, orientation, soiling and grid availability. Actual results will differ from this projection.